Problem 1: Consider the system $\dot{x} = -\sin(x)$. Prove that the origin is an asymptotically stable equilibrium point. (10 points)

$\dot{x} = 0 \qquad 0 = -\sin(x) \rightarrow x = k\pi$

$V(x) = \frac{1}{2}x^2 \rightarrow V(0) = 0,\ V(x) > 0$ when $x \neq 0$ positive definite

$\dot{V}(x) = \dot{x}x = x(-\sin(x)) = -x\sin(x)$ $\quad x\sin(x) > 0,\ x \neq 0$ $\quad \dot{V}(0) = 0$

$-x\sin(x) < 0,\ x \neq 0$

negative definite

$x = 0$ is asymptotically stable

Problem 2: For each of the following systems, using a form of quadratic Lyapunov function candidate to show that the origin is asymptotically stable. (30 points)

a) $\dot{x}_1 = -x_1 + x_2^2,\ \dot{x}_2 = -x_2$
b) $\dot{x}_1 = (x_1 - x_2)(x_1^2 + x_2^2 - 1),\ \dot{x}_2 = (x_1 + x_2)(x_1^2 + x_2^2 - 1)$
c) $\dot{x}_1 = -x_1 + x_1^2 x_2,\ \dot{x}_2 = -x_2 + x_1$

$V(x) = \frac{1}{2}(x_1^2 + x_2^2) \rightarrow V(0,0) = 0 \qquad V(x_1, x_2) > 0$ for $(x_1, x_2) \neq (0,0)$

positive definite

$\dot{V}(x) = x_1\dot{x}_1 + x_2\dot{x}_2$

a) $\dot{V}(x) = x_1(-x_1 + x_2^2) + x_2(-x_2) = -(x_1^2 + x_2^2) + x_1x_2^2$

$|(x_1, x_2)| < 1/2,\ |x_1| < 1/2$

$|x_1|\, x_2^2 \leq \frac{1}{2}x_2^2 \rightarrow \dot{V} \leq -x_1^2 - x_2^2 + \frac{1}{2}x_2^2 = -x_1^2 - \frac{1}{2}x_2^2$

asymptotically stable $\qquad (x_1, x_2) \neq (0,0) \rightarrow \dot{V} < 0$ negative definite

b) $\dot{V} = x_1(x_1 - x_2)(x_1^2 + x_2^2 - 1) + x_2(x_1 + x_2)(x_1^2 + x_2^2 - 1) = (x_1^2 + x_2^2)(x_1^2 + x_2^2 - 1)$

$x_1^2 + x_2^2 < 1 \rightarrow x_1^2 + x_2^2 - 1 < 0 \qquad (x_1, x_2) \neq (0,0),\ x_1^2 + x_2^2 > 0$

$\dot{V} = (x_1^2 + x_2^2)(x_1^2 + x_2^2 - 1) < 0$ $\qquad$ asymptotically stable

c) $\dot{V} = x_1(-x + x_1^2 x_2) + x_2(-x_2 + x_1) = -(x_1^2 + x_2^2) + x_1 x_2 + x_1^3 x_2$

$|x_1 x_2| \le \frac{1}{2}(x_1^2 + x_2^2)$ & $|(x_1, x_2)| < \delta < 1$

$|x_1^3 x_2| \le \frac{\delta^2}{2}(x_1^2 + x_2^2)$

$\dot{V} \le \frac{-1-\delta^2}{2}(x_1^2 + x_2^2) < 0$ asymptotically stable

Problem 3: Find out whether the following systems are controllable and observable (10 points)

$$\dot{x} = \begin{bmatrix} 0 & 4 & 0 & 0 \\ -1 & -4 & 0 & 0 \\ 5 & 7 & 1 & 15 \\ 0 & 0 & 3 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} u$$

a) $y = [1 \quad 0 \quad 0 \quad 0]x$
b) $y = [1 \quad 0 \quad 1 \quad 0]x$

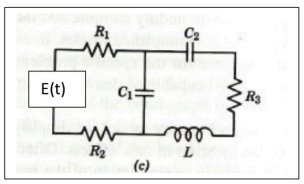
a) $C = [B \quad AB \quad A^2B \quad A^3B] = \begin{vmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 46 & -44 \\ 0 & 3 & -6 & 156 \end{vmatrix}$ rank(C) = 2 < 4 not controllable

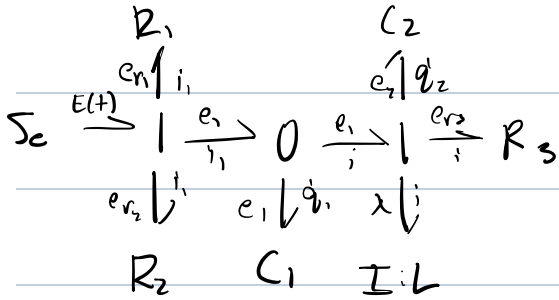
$O = \begin{vmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ -4 & -16 & 0 & 0 \\ 12 & 48 & 0 & 0 \end{vmatrix}$ rank(O) = 2 < 4 not observable

b) $O = \begin{vmatrix} 1 & 0 & 1 & 0 \\ 5 & 11 & 1 & 15 \\ -6 & -17 & 46 & -70 \\ 247 & 366 & -44 & 780 \end{vmatrix}$ rank(O) = 4 not controllable observable

Problem 4: For the following electric circuit, we $R_1=1$, $R_2=3$, $C_1=0.05$, $C_2=0.025$, $R_3=5$, $L=1$. The input to the system is a voltage source E(t). The output of the system is the voltage across C_1.

1) Model the system in a state space form (5 points);
2) Is the system stable? Controllable? (10 points)
3) Develop LQR controllers; compare the following cases
   a. Q is fixed, change R with two arbitrary values; (10 points)
   b. R is fixed, change one element of Q with two arbitrary values; (10 points)
4) Plot the result of the above four cases in time domain under $E(t)=0$ and there is 0.01 charge left in C_1. C_2 and L are initially at rest. Explain the results in your own words (10 points)





1) $\dot{q}_2 = 1/L\,\lambda$ $\quad \dot{q}_1 = i_1 - i = \frac{e_{r_1}}{R_1} - \frac{\lambda}{L} = \frac{1}{R_1+R_2}E(t) - \frac{1}{R_1+R_2}\frac{q_1}{C_1} - \frac{\lambda}{L}$

$\dot{\lambda} = e_1 - e_2 - e_{r_3} = \frac{q_1}{C_1} - \frac{q_2}{C_2} - \frac{R_3}{L}\lambda$

$$\begin{vmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \dot{\lambda} \end{vmatrix} = \begin{vmatrix} \frac{-1}{C_1(R_1+R_2)} & 0 & \frac{-1}{L} \\ 0 & 0 & 1/L \\ 1/C_1 & -1/C_2 & -\frac{R_3}{L} \end{vmatrix} \begin{vmatrix} q_1 \\ q_2 \\ \lambda \end{vmatrix} + \begin{vmatrix} \frac{1}{R_1+R_2} \\ 0 \\ 0 \end{vmatrix} E(t)$$

$$y = \begin{vmatrix} \frac{1}{C_1} & 0 & 0 \end{vmatrix} \begin{vmatrix} q_1 \\ q_2 \\ \lambda \end{vmatrix}$$

2) $$\dot{x} = \begin{vmatrix} -5 & 0 & 1 \\ 0 & 0 & 1 \\ 20 & -40 & -5 \end{vmatrix} x + \begin{vmatrix} 0.25 \\ 0 \\ 0 \end{vmatrix} u \qquad y = \begin{vmatrix} 20 & 0 & 0 \end{vmatrix} x$$

$$s^3 + 10s^2 + 85s + 200$$

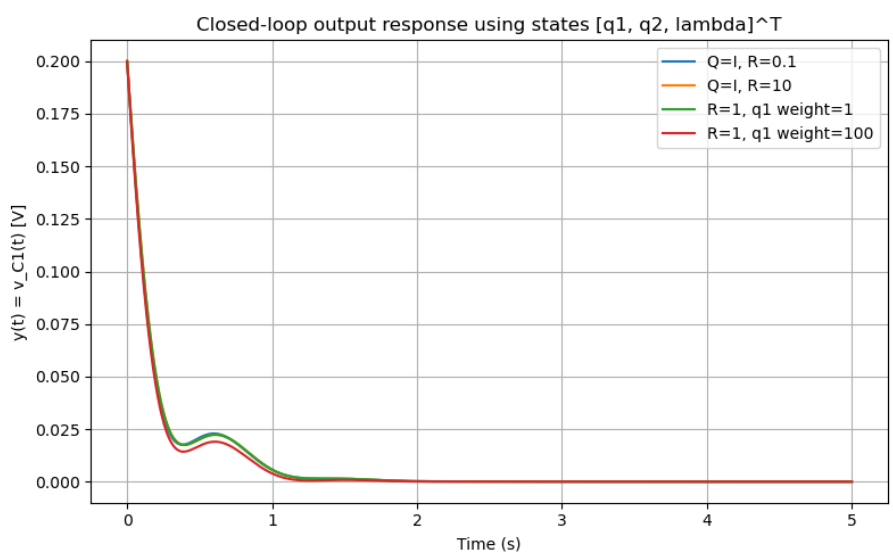
$$\lambda(A) \approx -3.154, \; -3.423 \pm j7.190$$

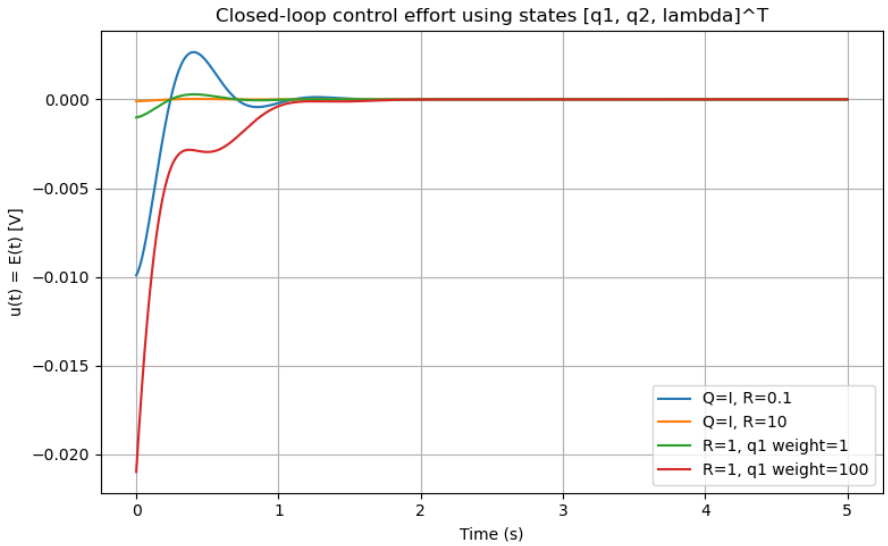
asymptotically stable

$$C = \begin{vmatrix} 0.25 & -1.25 & 1.25 \\ 0 & 0 & 5 \\ 0 & 5 & -30 \end{vmatrix} \qquad \text{rank}(C) = 3$$

controllable

3/4)





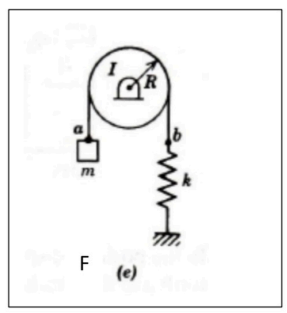
The plots show the closed-loop response of the capacitor voltage v_c1(t) and the corresponding control input u(t)=E(t) for different LQR weight selections. Since the system is already stable, all responses decay to zero with similar shapes, reflecting the natural dissipation of energy. However, the choice of weights affects how aggressively this energy is removed. Smaller R or larger weighting on q_1 results in larger initial control inputs and slightly faster decay of the output, while larger R produces minimal control effort and a slower, more passive response. Overall, the plots illustrate the tradeoff between control effort and performance, as well as how the weighting matrix Q can prioritize reducing specific states such as the capacitor voltage.

Problem 5: For the pully system, there is no slip between the roller and the string. Gravity can be neglected. A unit force is applied to the mass. The system parameters are:

$$m = 1, R = 1, I = 1, k = 1,$$

The displacement of the mass m is Y

1) Model the system in a state space form (5 points)
2) Develop LQI controllers to achieve the desired $Y = 1$ with 0 initial conditions and plot the system response (10 points)
3) Try two different Q matrices, compare the results in both time domain and frequency domain (10 points)



1)

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ \frac{-k}{m+\frac{I}{R^2}} & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ \frac{1}{m+\frac{I}{R^2}} \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x$$

plug in variables

$$\dot{x} = \begin{vmatrix} 0 & 1 \\ -\frac{1}{2} & 0 \end{vmatrix} x + \begin{vmatrix} 0 \\ \frac{1}{2} \end{vmatrix} u$$

2/3) python code